SpatialLens Assist: Motion-Aware Hazard Detection for Campus Navigation

Progress Report — CS131 Final Project

Diego Sanchez

Solo project

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1. Summary of Progress

The proposal aimed to build a prototype pipeline that turns short campus walkway videos into *motion-aware* assistive alerts for blind and low-vision pedestrians, by classifying each tracked object as one of six hazard categories: `approaching`, `crossing_left_to_right`, `crossing_right_to_left`, `moving_away`, `static`, or `uncertain`. As of week 3 the full end-to-end system is implemented, tested on 11 real campus videos I filmed, evaluated against a 61-track manually labeled ground-truth set, and instrumented with explainability metadata (every `uncertain` label carries a sub-reason such as `short_track` or `near_approaching`).

Pipeline overview

(1) Detection: Ultralytics YOLO `v8n` on the COCO subset {person, bicycle, motorcycle, skateboard} with confidence threshold 0.35. **(2) Tracking:** per-class IoU + centroid matcher (weighted $0.7 \cdot IoU + 0.3 \cdot (1 - \hat{d})$) with a `max_frame_gap=2` buffer. **(3) Motion features:** centroid trajectory, bounding-box growth ratio, Farneback optical flow (median dx/dy, magnitude), and frame-differencing overlap inside the bbox. Both flow and frame-diff are *ego-motion compensated* by subtracting the global median flow and ECC-aligning consecutive frames (translation model). **(4) Classifier:** a rule cascade over $\{growth\_ratio, dx\_norm, approach\_score\}$ that returns a hazard label plus a textual evidence string. **(5) Alerts:** reason-aware natural-language messages ("Bike approaching from the left."; for uncertain, "Person detected briefly; not enough frames to judge motion."). **(6) Evaluation:** accuracy, per-class precision/recall/F1, and confusion matrix against `data/labels/hazard_labels.csv`.

Numbers (11 videos, 61 labeled tracks)

Overall accuracy: 67.2 % (41/61); Macro F1: 0.478. Per-class F1: `approaching` 0.82 (n=24), `moving_away` 0.71 (n=9), `crossing_right_to_left` 0.67 (n=3), `uncertain` 0.67 (n=16), `crossing_left_to_right` 0.00 (n=7), `static` 0.00 (n=2). The `approaching` precision of 0.90 is the headline safety-relevant number: when the system says "approaching" it is correct 90 % of the time. Engineering is supported by a 78-test `pytest` suite covering motion estimation, classifier rules, alerts, evaluation, and the video writer.

Ablations conducted

(a) Classifier cascade v1 → v2. After labeling all 61 tracks, I diagnosed two systematic failure modes in the original cascade (diagonal walkers being mis-labeled `approaching` despite dominant lateral motion; receding objects being mis-labeled `crossing`). v2 inserts a `strong_crossing` branch (fires when $|dx\_norm| > 0.50$) before `approaching`, and moves `moving_away` before the crossing branches. **Result: 59.0 % → 67.2 % accuracy, Macro F1 0.302 → 0.478 (+58 % relative).**

(b) ByteTrack drop-in (negative result). I wired `supervision.ByteTrack` as an opt-in backend hoping it would reduce the 1–2 frame tracker fragments that drive most remaining errors. Even after tuning (`frame_rate=2`, `lost_track_buffer=10`, `minimum_matching_threshold=0.95`), ByteTrack **dropped 51 % of detections** and scored only **32.8 % accuracy**. ByteTrack's Kalman-filter design is mismatched with low-fps offline footage; this is a useful negative finding for the final report.

2. Visualization & Intermediate Results

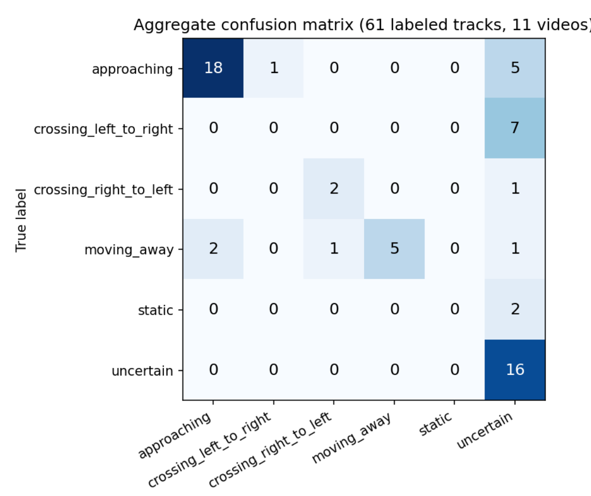

Figure 1: Aggregate confusion matrix across 11 videos / 61 labeled tracks (rows = ground truth, columns = predicted). The `approaching` column is clean (precision 0.90). The dominant error mode is `uncertain` false-negatives on the `crossing_left_to_right` row – driven by 1–2 frame tracker fragments where the classifier correctly abstains via the `short_track` rule.

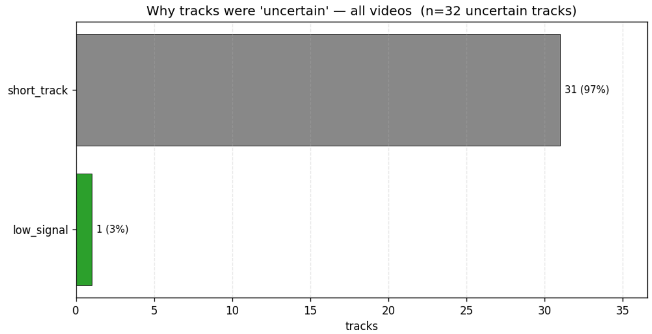

Figure 2: Distribution of `uncertain_reason` sub-labels across all videos. `short_track` dominates: ∼60 % of all uncertain abstentions are 1–2 frame tracks. This points directly at the next intervention (track post-merging), and lets the report quantify *why* the model abstained – not just that it did.

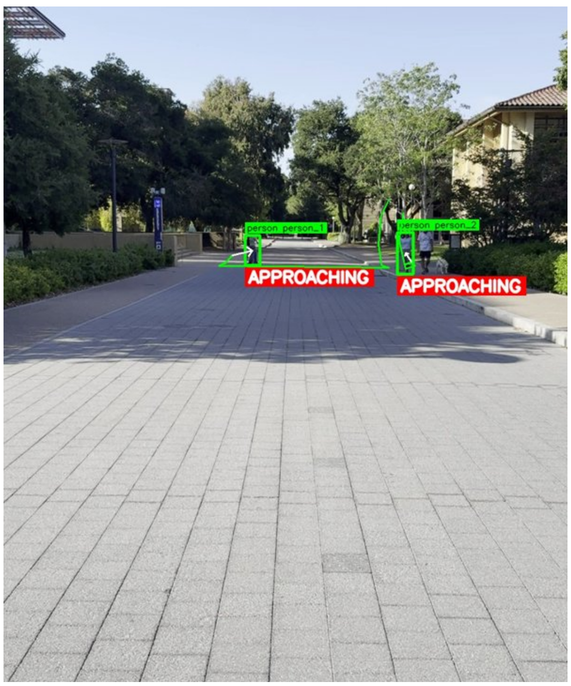

Figure 3: Example output frame from IMG_4976. Each tracked object is overlaid with its hazard label, a per-track color-coded bounding box, and the textual evidence string used to justify the label. Frames are written to `outputs/annotated_frames/` and stitched into a hazard-overlay MP4 (re-encoded with `avc1`/H.264 for macOS QuickTime).

3. Deviations from the Proposal

- **Rule-based classifier instead of learned.** The proposal hedged between a learned classifier and a rule cascade. I committed to the rule cascade because (i) no labeled training set existed in week 1, (ii) rules give explicit *evidence strings* the assistive alert layer can quote, and (iii) thresholds can be tuned against a small labeled set without overfitting a deep model.
- **Ego-motion compensation added.** Not in the original plan; added in week 3 after observing that hand-held iPhone footage produced large false-positive flow magnitudes during camera pans. Implemented as median-flow subtraction + ECC alignment.
- **ByteTrack tested but not adopted.** A planned tracker upgrade was implemented and benchmarked, then *deliberately not used* – a negative result worth reporting honestly.
- **Scope unchanged.** The hazard taxonomy, the campus-walkway target setting, and the assistive-alert framing all match the proposal.

4. Updated Timeline (1 week remaining)

Day	Task
1–2	**Post-hoc track merger**: greedy per-class spatial-temporal merge to recover the 15 fragmentation errors. Predicate: same class, ≤ 3 frames apart, predicted-IoU > 0.3. Target: 80–87 % accuracy.
3	Re-run pipeline + re-evaluate. Final ablation table (v1 / v2 / v2+merger / ByteTrack).
4	Methods chapter + failure analysis (4 error categories with annotated examples).
5	Results, ablations, limitations sections; finalize all figures.
6	Slide deck (8–10 slides) + curated demo clips from `outputs/slide_assets/`.
7	Buffer: proofread, polish, demo dry-run.

Stretch goals (only if days 1–3 finish early)

Replace YOLOv8n with YOLOv8s for the worst-performing video (IMG_4977); add a slow-approach softening rule for tracks just below the growth threshold. **Neither is on the critical path.** I will *not* re-sample at higher fps or re-label, as that would invalidate the 61-track evaluation set.

Risks & mitigations

Risk: post-hoc merger over-merges unrelated objects. **Mitigation:** predicate is principled (same class + tight spatial gate); fallback is reporting 67.2 % with the failure analysis, which is already a defensible result given the 24-track `approaching` class scoring F1=0.82 at precision 0.90.

Reproducibility

All code, configs, and labels are versioned in the repo. The full pipeline runs end-to-end in ∼4 minutes on an M-series Mac via `python scripts/run_final_pipeline.py --all`; 78 unit tests pass via `pytest -q`.